



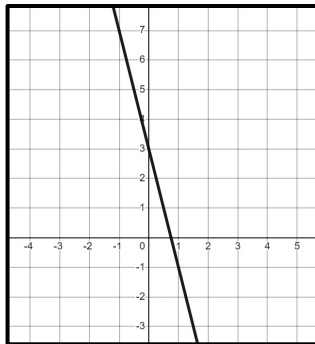
# The intercept point

## Task A: Identifying the y-intercept

1) For each question, circle the y-intercept on the graph and in the table.

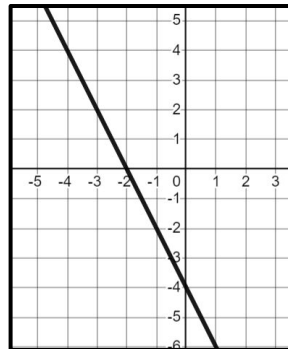
a)

|   |   |    |    |    |
|---|---|----|----|----|
| x | 0 | 1  | 2  | 3  |
| y | 3 | -1 | -5 | -9 |



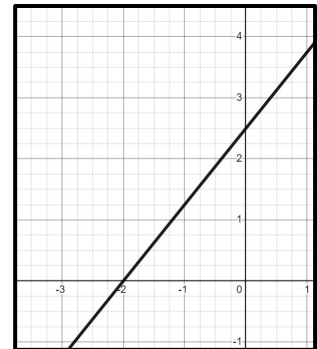
b)

|   |    |    |    |    |
|---|----|----|----|----|
| x | -2 | -1 | 0  | 1  |
| y | 0  | -2 | -4 | -6 |



c)

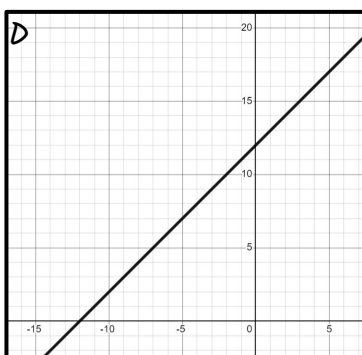
|   |    |     |   |     |
|---|----|-----|---|-----|
| x | -2 | 0   | 2 | 4   |
| y | 0  | 2.5 | 5 | 7.5 |



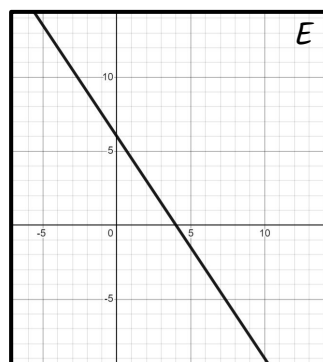
2) Write the y-intercept for each as a coordinate.

3) Match each y-intercept to a graph, a table and an equation.

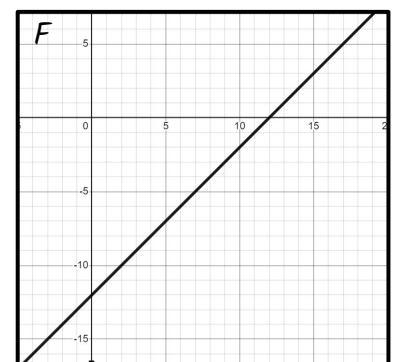
A (0,6)



B (0,12)



C (0,-12)



G

|   |   |   |   |
|---|---|---|---|
| x | 0 | 2 | 4 |
| y | 6 | 3 | 0 |

H

|   |     |    |    |
|---|-----|----|----|
| x | -10 | -5 | 0  |
| y | 2   | 7  | 12 |

I

|   |     |     |     |
|---|-----|-----|-----|
| x | -1  | 0   | 1   |
| y | -13 | -12 | -11 |

J  $y = x + 12$ K  $y + 12 = x$ L  $2y + 3x - 12 = 0$

**Task B: Calculating the  $y$ -intercept from tables of values**

For each table of values, work out the  $y$ -intercept.

a)

|     |   |      |    |    |    |
|-----|---|------|----|----|----|
| $x$ | 0 | .... | 2  | 3  | 4  |
| $y$ |   | .... | 13 | 16 | 19 |

b)

|     |   |      |    |    |    |
|-----|---|------|----|----|----|
| $x$ | 0 | .... | 6  | 7  | 8  |
| $y$ |   | .... | 19 | 21 | 23 |

c)

|     |   |      |    |    |    |
|-----|---|------|----|----|----|
| $x$ | 0 | .... | 25 | 26 | 27 |
| $y$ |   | .... | 17 | 21 | 25 |

d)

|     |   |      |    |    |     |
|-----|---|------|----|----|-----|
| $x$ | 0 | .... | 10 | 11 | 12  |
| $y$ |   | .... | -3 | -8 | -13 |

e)

|     |    |    |    |      |   |
|-----|----|----|----|------|---|
| $x$ | -5 | -4 | -3 | .... | 0 |
| $y$ | 15 | 12 | 9  | .... |   |

f)

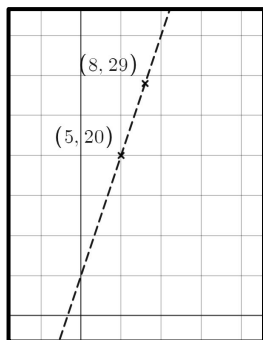
|     |     |     |     |
|-----|-----|-----|-----|
| $x$ | -12 | -11 | -10 |
| $y$ | -21 | -15 | -9  |



## Task C: Calculating the y-intercept from two coordinates

1) For each pair of coordinates, work out the y-intercept for the line which passes through them.

a) (5, 20) and (8, 29)



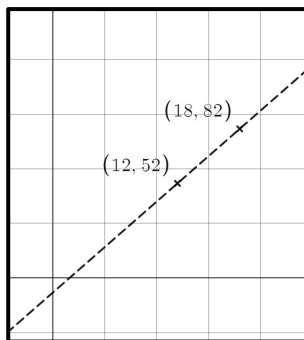
|   |   |
|---|---|
| 3 | 9 |
| 1 |   |

gradient  
=

|   |   |       |    |       |    |
|---|---|-------|----|-------|----|
| x | 0 | ..... | 5  | ..... | 8  |
| y |   | ..... | 20 | ..... | 29 |

y-intercept = ( \_ , \_ )

b) (12, 52) and (18, 82)



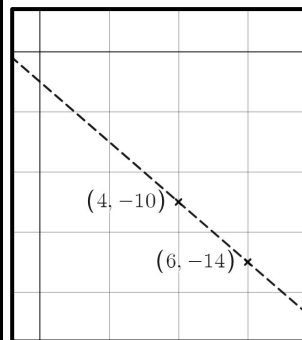
|   |  |
|---|--|
| 6 |  |
| 1 |  |

gradient  
=

|   |   |       |    |       |    |
|---|---|-------|----|-------|----|
| x | 0 | ..... | 12 | ..... | 18 |
| y |   | ..... | 52 | ..... | 82 |

y-intercept = ( \_ , \_ )

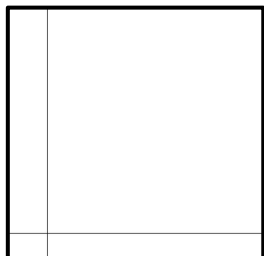
c) (4, -10) and (6, -14)



|   |    |
|---|----|
| 2 | -4 |
|   |    |

y-intercept = ( \_ , \_ )

d) (10, 8) and (18, 0)



|  |  |
|--|--|
|  |  |
|  |  |

gradient =

|   |  |  |    |       |    |
|---|--|--|----|-------|----|
| x |  |  | 10 | ..... | 18 |
| y |  |  |    | ..... |    |

y-intercept = ( \_ , \_ )

e) (7, 54) and (10, 75)

|   |  |  |   |       |    |
|---|--|--|---|-------|----|
| x |  |  | 7 | ..... | 10 |
| y |  |  |   | ..... |    |

y-intercept = ( \_ , \_ )

f) (6, 13) and (9, 22)

y-intercept = ( \_ , \_ )

